

They are capable of using Felica i-mode mobile wallet service and can be used for making purchases and subsequent payments in numerous shops equipped with a Felica card reader. The payment amount is directly billed to the customer's existing credit card account.

The current balance and payment records can be viewed using the mobile handset and the prepaid users can track the remaining balance and take time to reload the account with cash as and when needed. Also, during any payment transaction, the mobile subscribers can increase the credit limit at any service-supported store.

Such mobile payment services are widely popular in Japan and Shanghai and Beijing in China.

PayPal. PayPal, a wholly owned subsidiary of popular e-commerce auction site eBay, allows customers to make online payments during online purchases without using their credit cards. In order to use PayPal service, the user first has to create a PayPal account and deposit money in the account either using electronic debit from a bank account or by using a credit card. At the time of making online purchase, the user enters his PayPal account number along with the amount of payment and the recipient (vendor) details. PayPal transfers the amount from the user's PayPal account to the recipient either by cheque, electronically depositing to the recipient's bank account or, if the recipient is also a PayPal account holder, depositing in his PayPal account.

In order to allow payments through mobile devices, PayPal has introduced PayPal Mobile service, where PayPal facilities are extended to mobile phone users. To avail this service, the mobile handset user, who already has a PayPal account, needs to sign in by registering his mobile phone number and create a PIN. At the time of making payment through the mobile, the user sends an SMS to PayPal and types the PIN, amount and name of the recipient/vendor. PayPal then contacts the mobile user to confirm the payment



Near-field communication (NFC) technology

and completes the transaction. Such SMS-based mobile payment, where the user need not use his credit card, has become widely popular worldwide.

PayCircle. PayCircle is a global consortium that aims to provide mobile device users worldwide a standard means of making mobile payments, regardless of the payment systems used by the merchants or service providers. It was initially formed by computer and telecommunication giants such as Hewlett-Packard, Lucent, Siemens, Oracle and Sun Microsystems. The primary goal of PayCircle was to increase consumer use of m-Commerce by making mobile payments simple, uniform and secure. Membership of PayCircle is open to groups or companies involved in the mobile payment market, including banks, credit card companies, merchants or content providers.

PayCircle's main objective is to develop uniform application pro-

gramming interfaces (APIs) to facilitate 'open' payments. To provide a standardised payment infrastructure for mobile communication services, PayCircle developed some specifications and standardised procedures that support interaction between payment procedures and individual mobile communication services. PayCircle makes the specifications available to all interested parties and tests the APIs in multiple environments to drive their commercial application.

Mobipay. Mobipay system operating in Spain is a joint venture of all major Spanish mobile operators such as Telefonica Moviles, Vodafone and Amena and the Spanish financial institutions such as BBVA and SCH. Its objective is to launch mobile payment facilities by transforming the mobile handset to a user-friendly, flexible and safer device for micro payment. With Mobipay, the mobile handset users